CS412 USER MODELING

MIDTERM PROJECT REPORT

From User Interaction Data to User Segments and Adaptive User Models

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1. INTRODUCTION

This project analyzes interaction data from 500 users of a narrative educational game-based learning platform. Each user participated in a 30-minute session, with observations captured every 15 seconds, resulting in 120 observations per user and 60,000 total interaction records.

The primary objective is to answer the following research question: *What meaningful types of users can be identified from their behavior, affective states, and performance while using the game-based learning platform, and how could the system adapt to these different users?*

This analysis demonstrates the progression from raw interaction data to actionable user models: Interaction Data → User Features → User Segmentation → Segment Interpretation → Personas → System Adaptation.

Metric	Value
Total Users	500
Observations per User	120 (every 15 seconds)
Session Duration	30 minutes
Total Records	60,000
Observation Interval	15 seconds

2. UNDERSTANDING THE USER DATA

The dataset contains multiple types of information about each user's interaction with the game-based learning platform. These variables can be categorized as follows:

Behavioral Data:

• Screen State: Reading, Answering, or Exploring activities
• Clicks per 15-second interval: Interaction frequency
• Activity transitions: Switching between different activities

Affective Data:

Seven emotional states captured via facial recognition:
• Bored, Confused, Engaged, Focused, Frustrated, Happy, Neutral

Performance Data:

• Answer Outcome: Correct or Incorrect responses
• Response Time: Time taken to answer questions
• Narrative Progress: Percentage completion through the game

Important Distinction: Variables like User_ID and Capture_No identify observations but do not describe user behavior. These are excluded from clustering. Variables like Answer_Outcome and Response_Time are only present during answering activities (by design), appearing in approximately 31% of observations.

3. DATA PREPARATION AND FEATURE ENGINEERING

The event-level data (60,000 observations) was transformed into user-level features where one row represents one user (500 rows total). This transformation involved aggregating and summarizing each user's 30-minute interaction session.

Feature Categories Created:

Category	Features	Count
Behavioral	Proportions of time in Reading, Answering, Exploring	2*
Affective	Proportions of 6 emotional states	6*
Performance	Accuracy, avg response time, correct/incorrect counts	4
Interaction	Average clicks during each activity type	3
Engagement	Progress, transitions, positive/negative affect	4
	TOTAL SELECTED FOR CLUSTERING	**19**

*** Redundancy Removal:** From the initial 25 features created, 6 were removed to avoid multicollinearity. For example, since activity proportions sum to 1.0, we dropped 'Exploring' and kept 'Reading' and 'Answering'. Similarly, one affective state was dropped as they sum to 1.0.

Preprocessing: All 19 features were standardized using StandardScaler (mean=0, std=1) to ensure all features contribute equally to the clustering algorithm, regardless of their original scales.

4. USER SEGMENTATION METHOD

Clustering Technique: K-Means clustering was applied to identify groups of similar users.

Determining Optimal k: We evaluated cluster numbers from k=2 to k=10 using three metrics:

k	Silhouette Score	Davies-Bouldin Index	Assessment
2	0.314	1.306	Too simplistic (only 2 types)
3	0.283	1.405	One cluster dominates (64%)
4	**0.179**	**1.889**	**SELECTED - Well balanced**
5+	Decreasing	Increasing	Over-segmentation

Decision: k=4 Clusters

The selection of k=4 balances statistical validity with practical interpretability. While k=2 showed the best silhouette score, it provides insufficient granularity for adaptive system design. The k=4 solution creates well-distributed segments (16%, 29%, 20%, 35%) with distinct, actionable characteristics for each user type.

5. RESULTS AND USER SEGMENT PROFILES

Four distinct user segments were identified through clustering analysis. Each segment exhibits unique behavioral patterns, affective states, and performance characteristics.

SEGMENT 1: MARCUS - Struggling Readers (80 users, 16%)

Metric	Value	Rank
Reading Time	47.3%	HIGHEST
Accuracy	55.3%	LOWEST
Boredom Rate	29.2%	HIGHEST
Positive Affect	30.8%	LOWEST
Response Time	8.52s	SLOWEST
Progress	89.9%	LOWEST

Interpretation: Marcus represents learners who invest significant effort in reading but struggle with comprehension. This leads to low confidence, high disengagement (boredom), and the slowest progress through the narrative. The high reading time paired with low accuracy suggests reading comprehension is the primary bottleneck for this segment.

SEGMENT 2: SOPHIA - Confident Learners (147 users, 29%)

Metric	Value	Rank
Accuracy	86.1%	HIGHEST
Response Time	7.59s	FASTEST
Positive Affect	71.0%	HIGHEST
Answering Time	35.5%	HIGHEST
Correct Answers	36.3	MOST
Progress	93.6%	HIGHEST

Interpretation: Sophia represents high-achieving, confident learners who master content quickly and efficiently. They spend the most time actively answering questions, respond quickly and accurately, and maintain high positive emotional states throughout. This segment risks becoming bored if content lacks sufficient challenge.

SEGMENT 3: JORDAN - Struggling Answerers (98 users, 20%)

Metric	Value	Rank
Accuracy	47.2%	LOWEST
Confusion	25.5%	HIGHEST
Frustration	19.4%	HIGHEST
Negative Affect	55.0%	HIGHEST

Incorrect Answers	22.8	MOST
Response Time	8.82s	SLOWEST

Interpretation: Jordan represents persistent learners who attempt many questions (36% answering time) but struggle with accuracy. Despite their effort, they experience the highest rates of confusion and frustration, creating a negative emotional spiral. The pattern suggests they lack foundational understanding needed to answer correctly.

SEGMENT 4: ALEX - Balanced Explorers (175 users, 35%)

Metric	Value	Rank
Exploring Time	35.9%	HIGHEST
Accuracy	72.7%	Moderate-Good
Positive Affect	57.5%	Moderate
Negative Affect	30.2%	Moderate
Progress	92.6%	Good
Response Time	8.07s	Average

Interpretation: Alex represents thoughtful, balanced learners who take an exploratory approach to the game. They distribute time across all activities, with the highest exploration rate. Their moderate-to-good accuracy and balanced emotional state suggest they value understanding and discovery over speed or efficiency.

6. USER PERSONAS

Evidence-based personas were developed for each segment to humanize the data and guide system adaptation design. Each persona is grounded in actual cluster characteristics.

PERSONA 1: Marcus (Struggling Readers)

Typical Behavior: Spends 47% of time reading but achieves only 55% accuracy. Slow responses (8.52s) suggest he struggles to understand material.

Emotional State: 29% bored (highest), 52% negative affect overall. Only 31% positive emotions. Predominantly discouraged and disengaged.

Pain Points: Cannot understand reading material effectively, feels confused and frustrated, low confidence in answers, frequent failures.

System Needs: Scaffolding for reading comprehension, gradual difficulty progression, encouragement and positive feedback, alternative explanations, shorter reading segments.

PERSONA 2: Sophia (Confident Learners)

Typical Behavior: Balanced approach (38% reading, 36% answering, 27% exploring). Fastest responses (7.6s), 86% accuracy. Strategic and efficient.

Emotional State: 71% positive affect (highest). Rarely confused (9.7%) or frustrated (5.7%). Emotionally stable and confident.

Pain Points: May feel unchallenged by current difficulty, could become bored if content lacks adequate challenge.

System Needs: Advanced or challenging content, recognition and achievement milestones, faster pace, opportunities to tackle harder problems.

PERSONA 3: Jordan (Struggling Answerers)

Typical Behavior: High answering time (36%) despite low accuracy (47%). Most incorrect answers (22.8). Slowest responses (8.82s).

Emotional State: Highest confusion (25.5%) and frustration (19.4%). 55% negative affect (highest). Repeated failures create negative spiral.

Pain Points: Gets answers wrong repeatedly despite effort, feels confused by questions, frustration from lack of progress, may feel incapable.

System Needs: Immediate detailed feedback, guided worked examples, hints before answering, foundational content review, success experiences with easier problems.

PERSONA 4: Alex (Balanced Explorers)

Typical Behavior: Highest exploration (36%), well-balanced across activities. Moderate-good accuracy (73%). Curious and measured approach.

Emotional State: Balanced - 58% positive, 30% negative. Occasionally confused (12%) or bored

(11%) but not excessively. Healthy emotional state.

Pain Points: May not be challenged enough for mastery, exploration might not optimize efficiency, lacks push to move from competent to excellent.

System Needs: Content rewarding exploration and discovery, optional depth tracks, gentle challenge encouraging skill growth, narrative elements making learning meaningful.

7. PROPOSED SYSTEM ADAPTATIONS

For each segment, we propose specific system adaptations following the framework: Evidence → User Need → System Adaptation. These are not generic recommendations but targeted interventions based on observed patterns.

Adaptations for Marcus (Struggling Readers)

1. Comprehension Scaffolding: Highlight key concepts, provide 'explain this' buttons for complex paragraphs, insert mini-comprehension checks during reading, use visual aids alongside text.

2. Progressive Difficulty: Start with easier questions (60-70% target accuracy vs 86% for Sophia), gradually increase difficulty with success, celebrate consecutive correct answers.

3. Alternative Explanations: After incorrect answers show explanation plus worked example, provide concepts in multiple formats (text, visual, video, simulation), link to prerequisite content.

Adaptations for Sophia (Confident Learners)

1. Advanced Content: Unlock 'Expert Mode' at 80%+ accuracy, provide bonus advanced narrative branches, offer synthesis questions applying concepts across topics.

2. Achievement Recognition: Provide badges (Accuracy Streak, Speed Demon, Perfect Score), display skill rankings, offer expert certificates, narrative recognition of competence.

3. Personalized Pacing: Optional time-limited challenges, choice of content sequence, self-directed exploration, skip explanations if confident, custom difficulty settings.

Adaptations for Jordan (Struggling Answerers)

1. Intelligent Feedback: Show correct answer immediately, explain WHY answer is correct step-by-step, provide worked example with same structure, link to foundational concepts that led to error.

2. Hints & Scaffolding: Optional hint button guiding without revealing answer, scaffold complex questions into sub-questions, 'check my thinking' before submission, show similar solved examples first.

3. Adaptive Difficulty: Start with foundational/prerequisite content review, begin with easier versions of problem types, increase difficulty only after 3 consecutive correct, celebrate progress ('You got 2 right in a row!').

Adaptations for Alex (Balanced Explorers)

1. Enriched Exploration: Bonus narrative branches with no wrong answers, interactive simulations of concepts, 'dig deeper' sections on topics of interest, easter eggs rewarding thorough exploration.

2. Optional Depth Tracks: After each concept: 'Learn more' or 'Move on' choices, optional Depth Badges for topic mastery, curated collections of related content, optional creative projects.

3. Gentle Challenge: Stretch goals for slightly harder problems, reflection prompts ('Try explaining this to a friend'), optional time challenges without penalties, encouragement like 'You're close to mastery'.

8. LIMITATIONS, ETHICS, AND CRITICAL REFLECTION

User models are powerful but imperfect tools. This section addresses the limitations of our analysis and ethical considerations for implementation.

Affective State Classification Limitations

Facial expressions are culturally variable and context-dependent. The same expression can mean different things: a smile might indicate confusion (nervous humor), a frown might indicate deep concentration. Classifiers can be biased across demographics (skin tone, age, gender).

Risk: Misclassification leads to wrong support. If Marcus is misclassified as 'bored' when actually confused, the system provides wrong intervention.

Mitigation: Use affect as one signal among many, combine with behavioral data, allow self-reporting, audit classifier performance across demographics, provide override options.

Segmentation Limitations

Clusters represent AVERAGES, not perfect categories. Within-cluster variation exists and can be significant. Marcus's cluster accuracy ranges from 40-70%, not uniformly 55.3%.

Risk: Stereotyping users based on cluster membership, missing individual differences, premature labeling affecting growth mindset.

Mitigation: Use clustering as starting point, implement periodic re-clustering, track individual change, allow individuation, focus on growth rather than fixed labels.

Privacy Concerns

Facial data is biometric and uniquely identifying. Behavioral data reveals learning challenges that could affect student opportunities if leaked. Emotional state data is intimate and could be used to manipulate rather than support.

Mitigation: Minimize data collection, encrypt in transit and at rest, delete facial images after processing, limit access, transparency reports, clear deletion policies, informed parental consent for minors, alternative pathways to opt out.

Bias and Fairness

Facial recognition systems show documented bias (higher error rates on darker skin tones, different genders, ages). Clusters may correlate with demographic groups rather than just learning styles. Different adaptations may benefit some groups more than others.

Mitigation: Test classifier and clustering for bias across demographics, ensure training data represents diverse populations, audit whether all groups benefit equally from adaptations, provide diverse adaptation options, regular fairness audits.

9. CONCLUSION

This analysis demonstrates the power of user modeling for educational technology. Through systematic analysis of 60,000 interaction records from 500 users, we identified four distinct learner types with fundamentally different needs:

• **Marcus (16%)** - Struggling readers who need scaffolding and confidence building
• **Sophia (29%)** - Confident learners who need challenge and recognition
• **Jordan (20%)** - Struggling answerers who need feedback and foundational support
• **Alex (35%)** - Balanced explorers who need enrichment and gentle challenge

Each segment requires specific, targeted adaptations. A one-size-fits-all approach will fail for the majority of users.

Key Insights:

1. Emotional state matters as much as performance metrics
2. Practice alone is insufficient - feedback quality matters
3. Different users require fundamentally different support strategies
4. Within-cluster variation exists - models are starting points, not final labels

Ethical Responsibility:

User models are REPRESENTATIONS of people, not perfect descriptions. Implementation must prioritize: (1) Transparency - students understand system choices, (2) Autonomy - students can override decisions, (3) Consent - informed agreement for data collection, (4) Monitoring - track whether adaptations help or harm, (5) Fairness - test for disparate impact across demographics.

The goal is not perfect prediction, but respectful, beneficial, transparent support for diverse learners.

Students are individuals first, cluster members second.

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